

"This property belongs to the half-blood princess UvD in collaboration with AI."

- **25 mcq s from applying real world examples to lecture content**
- **25 mcq s from covering key topics of lecture content**
- **Cloud Security, Monitoring, Databases, Analytics Services, Cloud Migrations, Cloud Native Applications, and Containerization.**

25 mcq questions from Lec 6-11

incorporating real-world examples to provide context and make the concepts more applicable to practical situations.

Cloud Security and Monitoring

1. ****A financial services company uses Multi-Factor Authentication (MFA) to access sensitive data. Which of the following is a common second factor in MFA used alongside a password?****

- A. Social Security Number
- B. Security Question
- C. Biometric Scan (e.g., fingerprint)
- D. Username
- ****Answer:** C (Biometric factors like fingerprints or facial recognition are often used as second factors in MFA)**

2. ****A healthcare company is storing patient records in the cloud and wants to prevent unauthorized access to this data. Which IAM principle should they apply to restrict user access strictly to necessary information?****

- A. Role-Based Access Control (RBAC)
- B. Multi-Factor Authentication (MFA)
- C. Principle of Least Privilege (PLOP)
- D. Open Access
- ****Answer:** C (Applying the Principle of Least Privilege minimizes access to sensitive data)**

3. ****A large e-commerce website uses Amazon CloudWatch to monitor server performance. If traffic spikes suddenly during a sale, which type of alert is most useful to notify administrators?****

- A. Heartbeat Alert

- B. Anomaly Detection Alert
- C. Time-Based Alert
- D. Weekly Report Alert
- **Answer:** B (Anomaly detection alerts can identify unexpected traffic spikes)

4. **A company implements Software-Defined Security (SDS) to manage network security automatically. What benefit does SDS provide in a cloud environment?**

- A. It provides fixed, static firewall configurations.
- B. It allows dynamic policy changes and automatic threat response.
- C. It encrypts all data at rest.
- D. It only monitors outbound traffic.
- **Answer:** B (SDS enables automated, software-driven security changes)

5. **A tech startup uses Datadog to monitor its cloud applications. What is a primary benefit of using Datadog for cloud monitoring?**

- A. It serves as a database system for the startup's data.
- B. It provides visibility into system metrics like CPU usage and latency.
- C. It automates data backups.
- D. It only offers real-time alerts.
- **Answer:** B (Datadog monitors and reports on system metrics)

Cloud Database and Analytics Services

6. **A gaming company uses a graph database to track player connections and interactions. Which database type is this?**

- A. Key-Value Store
- B. Document Store
- C. Columnar Database
- D. Graph Database
- **Answer:** D (Graph databases are optimized for managing interconnected data)

7. **An online retailer needs to store transaction records with strict data integrity and fast processing for high volumes of orders. Which database type should they choose?**

- A. NoSQL
- B. OLAP Database
- C. Key-Value Store
- D. OLTP Database
- **Answer:** D (OLTP is best for transaction processing with high integrity and speed)

8. **A social media app stores user sessions in a NoSQL database to scale with a dynamic schema. Which type of NoSQL database is suitable for this use case?**

- A. Key-Value Store
- B. Document Store
- C. Graph Database
- D. Relational Database
- **Answer:** B (Document stores handle flexible, unstructured data formats like JSON)

9. **A financial analytics company uses a data warehouse for aggregating and analyzing transaction data. What is a benefit of using a data warehouse for this purpose?**

- A. Storing raw data for analysis
- B. Enabling real-time transaction processing
- C. Providing structured, historical data for business intelligence
- D. Optimizing data for machine learning algorithms only
- **Answer:** C (Data warehouses are used for storing and analyzing structured historical data)

10. **A fast-food chain uses Redis to cache menu data, providing quick access during peak hours. Why might Redis be chosen over Memcached for this purpose?**

- A. It supports non-persistent caching only.
- B. Redis is non-volatile and can persist data through reboots.
- C. Memcached is too slow for caching.
- D. Redis cannot support multiple data structures.

- **Answer:** B (Redis can persist data, making it suitable for quick-access caching needs)

Cloud Migrations

11. **A retail company moves its e-commerce website “as is” to AWS without modifying the application code. Which migration strategy does this represent?**

- A. Rehost
- B. Refactor
- C. Re-platform
- D. Rebuild

Answer: A (Rehosting, or lift-and-shift, involves moving an application to the cloud without changes)

12. **A company decides to migrate its HR system to Workday, a SaaS solution. What cloud migration approach does this describe?**

- A. Re-platform
- B. Retire
- C. Repurchase
- D. Refactor

Answer: C (Repurchase refers to moving to a SaaS product, like moving an HR system to Workday)

13. **A logistics company modifies its existing application to take advantage of cloud features like auto-scaling. What migration strategy does this involve?**

- A. Rehost
- B. Refactor
- C. Retain
- D. Repurchase

Answer: B (Refactoring involves modifying the application to benefit from cloud-native features)

14. **A company merges with another firm and faces compatibility issues with on-premises applications. Which migration challenge does this illustrate?**

- A. Cost of migration
- B. Shortage of skills
- C. Integration complexity
- D. Data redundancy
- **Answer:** C (Compatibility issues often arise during mergers, making integration complex)

15. **An enterprise follows a hybrid cloud model during migration to gradually move critical workloads to the cloud. What is a primary benefit of this approach?**

- A. Immediate cost savings
- B. Full dependency on cloud services
- C. Flexibility in using on-premises and cloud resources
- D. Complete data privacy
- **Answer:** C (Hybrid models allow flexible use of both on-premises and cloud resources)

Cloud Native Applications

16. **Netflix uses a microservices architecture to handle its large-scale, global streaming services. What is a major benefit of this approach?**

- A. Low latency for each microservice
- B. Independent scaling of individual services
- C. Simplified development for monolithic applications
- D. Reduced deployment frequency
- **Answer:** B (Microservices allow each service to scale independently)

17. **An airline develops a stateless microservices application for booking and ticketing. What is an advantage of using stateless microservices?**

- A. Improved data persistence across sessions
- B. Easier horizontal scaling without session management
- C. Increased dependency on specific hardware

- D. Complex to deploy and manage
- **Answer:** B (Stateless services allow easier scaling across multiple servers)

18. **Spotify follows the 12-factor app methodology in its cloud-native applications. Which principle supports this by advocating for one codebase used across multiple environments?**

- A. Configuration as Code
- B. Port Binding
- C. Build, Release, Run
- D. Codebase
- **Answer:** D (A single codebase is a fundamental 12-factor principle)

19. **Uber uses a cloud-native architecture to improve its app's reliability and scalability. What characteristic of cloud-native architecture supports rapid deployment and easy scaling?**

- A. Monolithic design
- B. Serverless and containerized microservices
- C. Manual testing and configuration
- D. On-premises data centers
- **Answer:** B (Serverless and containerized microservices support quick scaling and deployment)

20. **Airbnb implements continuous integration and continuous deployment (CI/CD) to streamline updates. Which benefit of CI/CD is most advantageous in this context?**

- A. Reduced test coverage
- B. Frequent and reliable code deployment
- C. Manual intervention required for every update
- D. Increased application size
- **Answer:** B (CI/CD enables frequent, reliable deployments, critical for modern apps)

Containerization and Infrastructure as Code (IaC)

21. **A fintech company chooses Docker to containerize its microservices, allowing seamless movement between cloud providers. What is a primary advantage of using Docker containers?**

- A. Limited portability
- B. Slower start-up time than V

Ms

- C. Resource efficiency and portability
- D. Dedicated operating system for each container
- **Answer:** C (Docker containers are portable and resource-efficient)

22. **A logistics firm uses Kubernetes to orchestrate containerized applications for scaling during peak delivery periods. Why is Kubernetes suitable for this task?**

- A. It only works with specific hardware.
- B. It automates container scaling and management.
- C. It requires fewer resources than Docker.
- D. It offers a built-in database for logistics data.
- **Answer:** B (Kubernetes automates container management and scaling)

23. **A tech company adopts Infrastructure as Code (IaC) to streamline its infrastructure provisioning. What is a primary benefit of IaC in this scenario?**

- A. Allows manual setup of infrastructure
- B. Simplifies and automates infrastructure configuration
- C. Limits environment replication
- D. Requires traditional data center architecture
- **Answer:** B (IaC automates infrastructure setup, reducing manual errors)

24. **An e-commerce platform replicates its production environment for testing using IaC. What advantage does this provide?**

- A. Simplifies manual updates across environments
- B. Allows identical testing and production environments

- C. Ensures environments are unique and hard to replicate
- D. Increases the time needed to provision environments
- **Answer:** B (IaC allows easy replication for consistent testing and production environments)

25. **A media company uses Terraform to manage its cloud infrastructure across multiple providers. What is Terraform classified as?**

- A. Configuration management tool
- B. Database management tool
- C. Containerization tool
- D. Cloud-agnostic Infrastructure as Code (IaC) tool
- **Answer:** D (Terraform is an IaC tool that is cloud-agnostic)

25 (MCQs) covering the key topics from the lecture materials on Cloud Security, Monitoring, Databases, Analytics Services, Cloud Migrations, Cloud Native Applications, and Containerization.

Cloud Security and Monitoring

1. **Which of the following principles restricts user access strictly to what they need to perform their tasks?**

- A. Principle of Complete Access
- B. Principle of Least Privilege
- C. Role-Based Access Control
- D. Open Access Policy
- **Answer:** B

2. **What is a common risk associated with weak access management in cloud environments?**

- A. Data redundancy
- B. Data integrity
- C. Account hijacking
- D. Increased availability

- **Answer:** C

3. **Which type of authentication requires users to provide multiple forms of verification, such as a password and a fingerprint?**

- A. Single Sign-On (SSO)
- B. Multi-Factor Authentication (MFA)
- C. Public Key Infrastructure (PKI)
- D. Role-Based Authentication

- **Answer:** B

4. **What is the main function of Software Defined Security (SDS) in cloud environments?**

- A. It automates security operations and applies policy-driven controls.
- B. It establishes static security protocols for applications.
- C. It is a backup strategy for cloud resources.
- D. It primarily manages encryption keys.

- **Answer:** A

5. **Which of these would likely be used for automated performance monitoring in a cloud environment?**

- A. Datadog
- B. Oracle DBMS
- C. Redis
- D. GitLab

- **Answer:** A

6. **What type of alert would notify administrators if a system heartbeat check is missed?**

- A. Threshold-based alert
- B. Heartbeat alert
- C. Anomaly detection-based alert

- D. Performance alert

- **Answer:** B

Cloud Database and Analytics Services

7. **Which database model would be ideal for managing a social network with highly interconnected user data?**

- A. Key-Value Store

- B. Relational Database

- C. Document Store

- D. Graph Database

- **Answer:** D

8. **In the CAP theorem, if a database prioritizes consistency and availability, which property might it sacrifice?**

- A. Partition tolerance

- B. Consistency

- C. Availability

- D. Scalability

- **Answer:** A

9. **What is a primary benefit of using a data warehouse over a data lake for business intelligence (BI) purposes?**

- A. Holds unprocessed data

- B. Stores raw data for data science

- C. Provides structured and filtered data for BI

- D. Ideal for storing unstructured data

- **Answer:** C

10. **Which type of database workload is characterized by rapid, transactional data processing and short response times?

- A. Online Analytical Processing (OLAP)
- B. Online Transaction Processing (OLTP)
- C. Document Processing
- D. Data Warehousing
- **Answer:** B

11. **Which non-relational database type is optimized for retrieving large amounts of data by columns?**

- A. Key-Value Store
- B. Document Store
- C. Columnar Database
- D. Graph Database
- **Answer:** C

12. **What caching system would be ideal for a shopping platform that needs sub-millisecond response times and persistence across reboots?**

- A. Memcached
- B. Oracle DB
- C. Redis
- D. PostgreSQL
- **Answer:** C

Cloud Migrations

13. **Which cloud migration strategy involves moving applications “as is” without modifying their code or architecture?**

- A. Re-platform
- B. Rehost
- C. Refactor
- D. Replace
- **Answer:** B

14. ****Which of the following is a challenge commonly faced during cloud migration?****

- A. Inability to scale
- B. Resistance to adapting new technology
- C. Unlimited data storage
- D. Reduced operational costs
- ****Answer:** B**

15. ****Moving a custom-built HR system to a SaaS solution like Workday is an example of which cloud migration strategy?****

- A. Refactor
- B. Repurchase
- C. Retire
- D. Re-platform
- ****Answer:** B**

16. ****Which migration strategy would be most suitable for a company looking to move their legacy application and take advantage of cloud-native features?****

- A. Rehost
- B. Re-platform
- C. Refactor
- D. Retain
- ****Answer:** C**

17. ****What is the main advantage of a “hybrid cloud” approach during migration?****

- A. Full dependency on public cloud services
- B. Utilizing on-premises and cloud resources during the transition
- C. Simplified management and cost savings
- D. Moving everything directly to the cloud without planning

- **Answer:** B

Cloud Native Applications

18. **Which architectural style is defined by breaking down applications into independent, modular services?**

- A. Layered Architecture
- B. Monolithic Architecture
- C. Microservices Architecture
- D. Client-Server Architecture

- **Answer:** C

19. **A company wants to deploy a stateless application across multiple servers to handle a variable workload. Which cloud-native approach is best?**

- A. Monolithic
- B. Multi-layered
- C. Stateless microservices
- D. Stateful microservices

- **Answer:** C

20. **Which company is known as a pioneer in cloud-native applications and operates on a microservices architecture?**

- A. Microsoft
- B. Netflix
- C. IBM
- D. Facebook

- **Answer:** B

21. **What does the 12-factor app methodology emphasize for modern applications?**

- A. Rigid configuration
- B. Building applications as monolithic structures

- C. Scaling stateless processes
- D. Storing configurations directly in the code
- **Answer:** C

Containerization and Infrastructure as Code (IaC)

22. **Which of the following is NOT a characteristic of containers?**

- A. Shares the host operating system's kernel
- B. Requires its own complete OS installation
- C. Faster start-up than virtual machines
- D. Uses fewer resources compared to VMs
- **Answer:** B

23. **Which tool is typically used for container orchestration to manage and scale containers across multiple hosts?**

- A. Docker
- B. Kubernetes
- C. Ansible
- D. Jenkins
- **Answer:** B

24. **What is the main purpose of Infrastructure as Code (IaC)?**

- A. To manually configure cloud infrastructure
- B. To automate infrastructure provisioning and management with code
- C. To secure databases in cloud environments
- D. To perform real-time data analytics
- **Answer:** B

25. **Which benefit does IaC provide when setting up multiple environments for testing and production?**

- A. Ensures each environment is unique and hard to replicate
- B. Allows each environment to be quickly duplicated with minimal errors
- C. Simplifies manual updates across all environments
- D. Limits flexibility in environment configurations
- ****Answer:** B**